Binomial Distribution

The **Binomial Distribution** is a **discrete probability distribution** that describes **the number of successes in a fixed number of independent experiments (trials)** where each trial has only **two possible outcomes**.

These outcomes are usually called **Success and Failure**

Example:
You toss a coin 4 times and success is getting head.

- Here each toss is independent experiment
- You can get
  - 0 success
  - 1 success
  - 2 success -->
  - 3 success
  - 4 success.

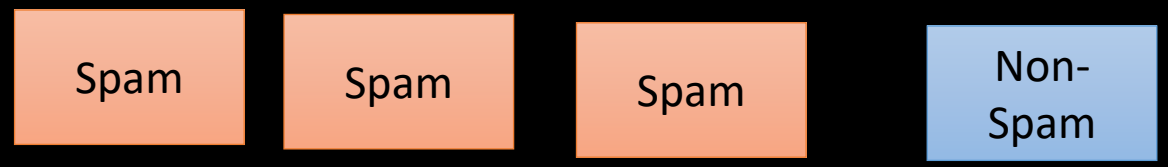
Binomial Distribution

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Example:

You pick 4 random emails from inbox and success is getting spam.

- Here picking each email is independent experiment
- You can get
  - 0 success -> no spam
  - 1 success -> 1 spam
  - 2 success -> 2 spam
  - 3 success - >3 spam
  - 4 success. - >4 spam



Binomial Distribution

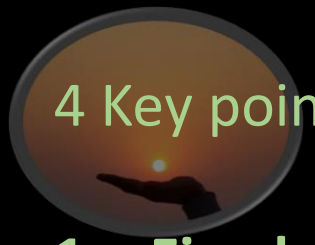
$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Example:

You pick 4 random items from an assembly line for inspection. Success is getting defective piece.

- Here picking item is an independent experiment
- You can get
  - 0 success -> no defect
  - 1 success -> 1 defect
  - 2 success -> 2 defect
  - 3 success - >3 defect
  - 4 success. - >4 defect

Defect	Defect	Defect	Non-Defect



4 Key point of Binomial distribution

1. **Fixed number of trials (n):** Number of experiments is predetermined.

2. **Only two outcomes:**
Each trial must result in Success (S) or Failure (F)

3. **Constant Probability:**
Probability of success remains same for every trial:

$$P(success) = p$$
$$P(failure) = q = 1 - p$$

4. **Independent trials:**
Outcome of one trial does not affect another.

Example: Coin tosses are independent.

Random Variable

Let:

$$X = \textit{Number of successes in } n \textit{ trials}$$

Then:

$$X \sim Binomial(n, p)$$

Where,

n= number of trials

p= probability of success

Number of ways to choose k successes from n trials:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Example: There are **6 ways** of getting 2 H in 4 coin tosses:
HHTT, HTTH, HTHT, TTHH, THTH, THHT

Here **Probability Mass Function (PMF)** is given by

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

This is the probability of k successes

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Suppose:

- Coin tossed 5 times
- Probability of head = 0.5

Question:

What is probability of getting exactly 3 heads?

$$n = 5, \quad p = 0.5, \quad k = 3$$

$$P(X = 3) = \binom{5}{3} (0.5)^3 (0.5)^2$$

$$= 10 \times (0.5)^5$$

$$= 0.3125$$

Example:
An assembly line produces machine parts. The probability that a part is **defective** is **0.2**.
Three parts are selected randomly and independently.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

What is the probability that exactly 2 parts are defective?

Solution:
Here we have 3 experiments: n=3
Success is getting a defective part. So k=2

Substitute values $(n = 3,\ k = 2,\ p = 0.2)$:

$$P(X = 2) = \binom{3}{2} (0.2)^2 (0.8)^1 \quad = 0.096$$

Code

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom # binomial distri
```

Code

You toss a coin 5 times and prob of success (getting H) is 0.5

What is the PMF ?

What is probability of getting 3 heads ?

```
# Number of trials (keep constant)
n = 5

# Possible number of successes
x = np.arange(0, n+1) # x = 0, 1, 2, 3, 4, 5

p = 0.5

probabilities = binom.pmf(x, n, p)
probabilities
```

```
array([0.03125, 0.15625, 0.3125 , 0.3125 , 0.15625, 0.03125])
```

Code

```
array([0.03125, 0.15625, 0.3125 , 0.3125 , 0.15625, 0.03125])
```

Probability of getting 0 heads = 0.03125

Probability of getting 1 heads = 0.15625

Probability of getting 2 heads = 0.3125

Probability of getting 3 heads = 0.3125

Probability of getting 4 heads = 0.15625

Probability of getting 5 heads = 0.03125

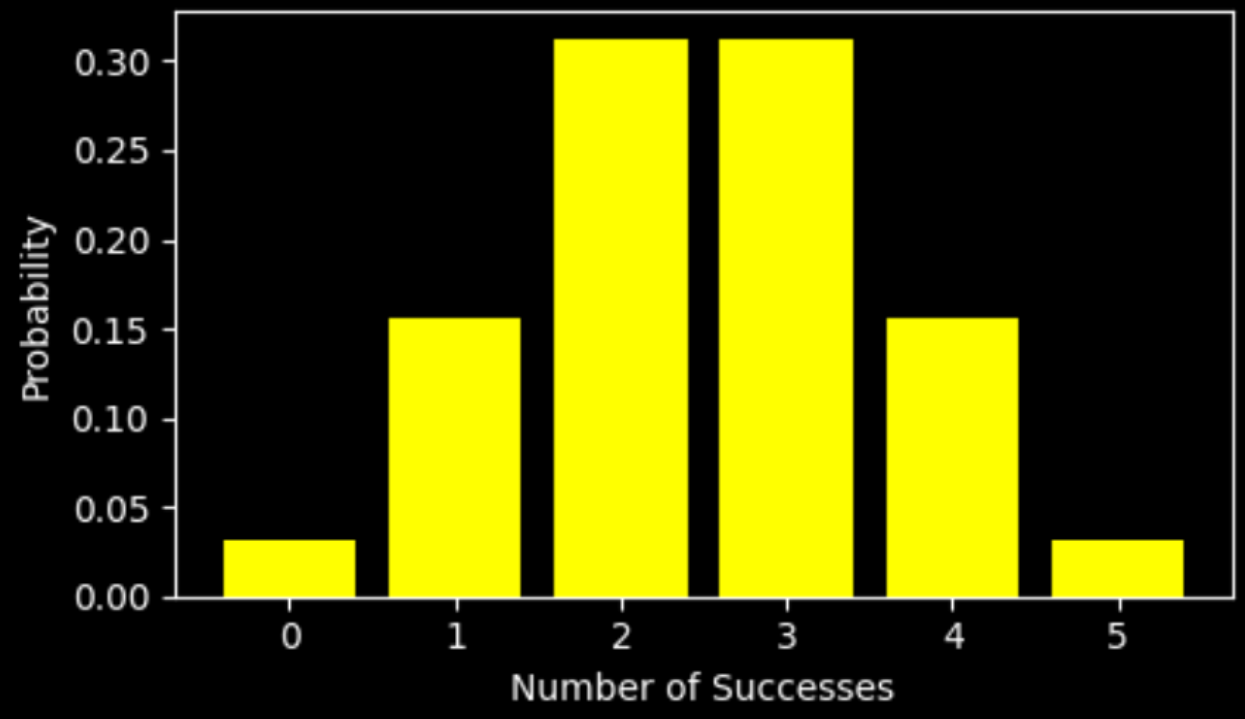
Code

```python
# bar plot:
plt.style.use("dark_background")
plt.figure(figsize=(5,3))

plt.bar(x, probabilities, color='yellow')

plt.xlabel("Number of Successes")
plt.ylabel("Probability")

plt.tight_layout()
plt.show()
```

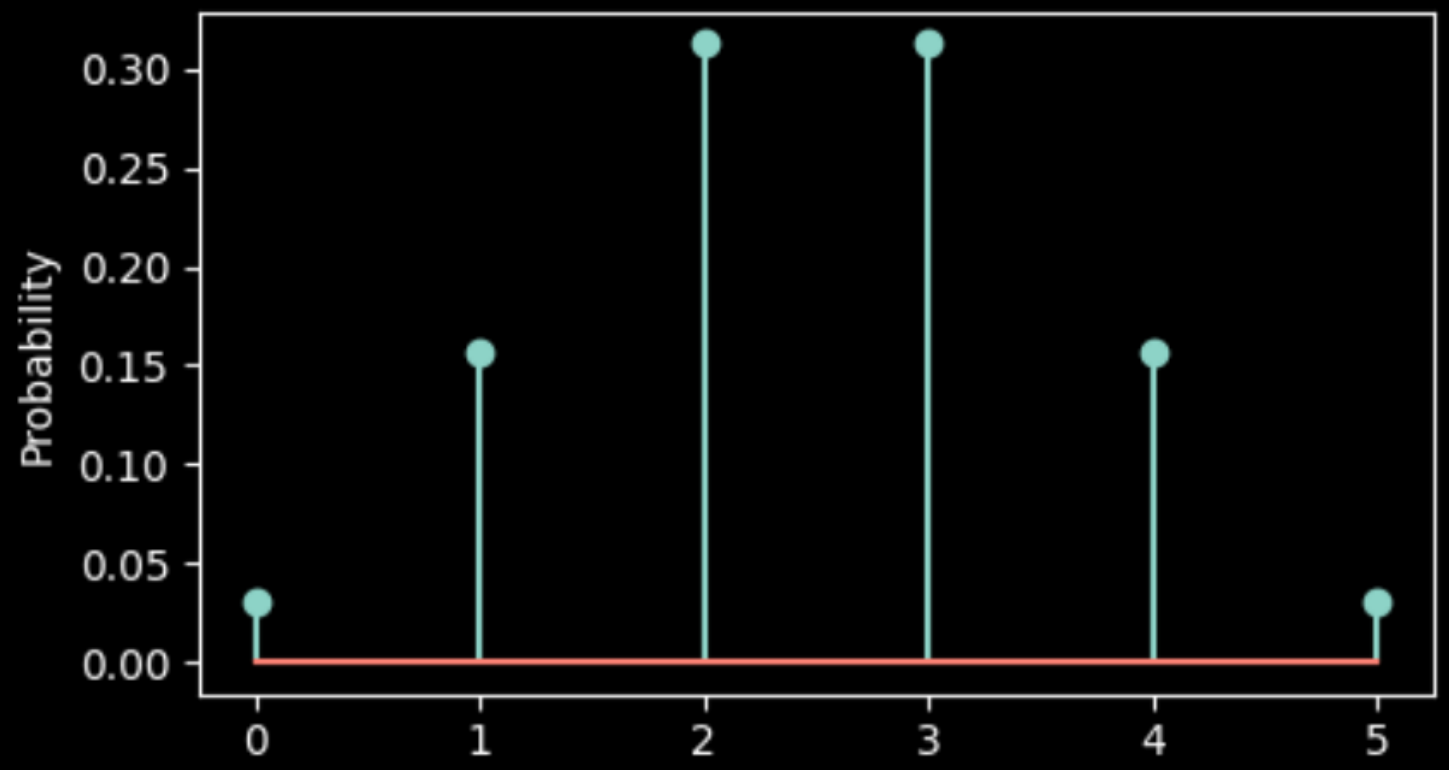


Code

```python
# Alternative: stem plot
plt.figure(figsize=(5,3))

plt.stem(x, probabilities)

plt.xlabel("Number of Successes")
plt.ylabel("Probability")
plt.tight_layout()
plt.show()
```



Real-Life Applications

- Binomial distribution is widely used in:
- Machine Learning classification accuracy
- Quality control testing
- A/B testing
- Medical trials

Mean (Expected Value)

$$E(X) = np$$

Variance

$$Var(X) = np(1-p)$$

Standard Deviation

$$\sigma = \sqrt{np(1-p)}$$

P(k success) = p^k

P(k failure) = $(1-p)^{n-k}$